

Besher Alkurdi

NLP Research Engineer



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  Jyväskylä, Finland (Relocatable)

Experience

- Research Engineer at Huawei** Aug 2023 - Aug 2024
- Enhanced custom in-house distributed training library with parameter-efficient methods, context expansion, and training stabilization techniques, achieving 8x resource reduction without performance degradation.
 - Conducted customer requirements analysis sessions, architected ML solutions for business needs, and defined PoC implementation roadmaps with cross-functional teams.
 - Coded LLM architectures enabling compatibility with standard libraries, delivering significant inference speedup.
 - Built scalable multi-backend inference platform with RESTful/streaming APIs, supporting scalable deployment and advanced sampling methods.
 - Wrote multilingual synthetic data generation toolkit and fine-tuning pipelines for customer LLM adaptations.
 - Ran ZeRO/FSDP/DeepSpeed multi-node training, tested models and supplied teams with improved data pipelines.
- Undergraduate Research Fellow at Yildiz Technical University** June 2022 - March 2023
- Curriculum learning research in *publicly funded project*, implementing cyclical and difficulty-based training strategies.
 - Developed data augmentation pipeline and experimented with auxiliary classifiers for early exit strategies.
 - Deployed distributed training systems using TensorFlow, PyTorch, and Horovod on lab computers.
 - Trained CLIP-style multimodal models with contrastive learning.
- Research Intern at New Mind AI** July 2021 - October 2021
- Conducted research on state-of-the-art natural language processing techniques and their application in a system assisting lawyers in lawsuits focusing on Document AI applications in the Turkish language.

Education

- MSc. in Artificial Intelligence** Thesis Phase
University of Jyväskylä
- BSc. in Computer Science** June 2023
Yildiz Technical University
2nd in class, GPA: 3.84/4

Publications

- Scaling Up Paraphrase Generation Datasets with MT and Semantic Similarity Filtering** 2024
- An extension of our prior work. Experimented with data augmentation using paraphrasing.
- Semantic Similarity Based Filtering for Turkish Paraphrase Dataset Creation** December 2022
- A method for creating paraphrase datasets from parallel corpora along the largest paraphrase datasets in Turkish.
- Linguistic-based Data Augmentation Approach for Offensive Language Detection** 2022
- Augment classification datasets with a linguistic-based approach improving the detection of offensive language.

Skills

ML/DL: PyTorch, TensorFlow, HuggingFace, OpenCV, NLP, TRAX, ZeRO, FSDP, DeepSpeed

Programming: Python, C, Rust, SQL, Java

Tools: Linux, Git/GitHub, ROS, Docker

Languages

Arabic: Native • **English:** Advanced • **Turkish:** Advanced • **Finnish:** Beginner • **French:** Beginner

Certificates

Deep Reinforcement Learning Course <i>HuggingFace</i> Q-Learning, PPO.	May 2024
Generative Adversarial Networks (GANs) Specialization <i>DeepLearning.AI</i> GAN components, data augmentation, privacy.	July 2023
Natural Language Processing Specialization <i>DeepLearning.AI</i> NLP, Transformers, Attention models, Machine Translation.	June 2023
Deep Learning Specialization <i>DeepLearning.AI</i> Deep learning, including GANs and VAEs.	February 2023
Machine Learning Specialization <i>DeepLearning.AI</i> ML fundamentals and practical applications.	August 2022
Introduction to Computational Thinking and Data Science <i>edX</i> Introductory course in data science and ML.	May 2021

Verification: All certificates are verifiable at <https://www.linkedin.com/in/mrbesher>.

Volunteer Experience

Volunteer Researcher at COSMOS AI Research Group	December 2024 – Present
Conducting research on improving the performance of large language and vision-language models using unlabeled data and reinforcement learning without reliance on stronger teacher supervision.	
Tutor at Schoolhouse.world	February 28, 2021 - May 20, 2021
Tutoring students worldwide in subjects such as Math, Computer Science, and more.	
Tutor at Aqsa Youth Institution	November 30, 2020 - August 31, 2021
Taught computer, networking, and programming basics to middle school and high school refugee students.	
Translator at Ask Me	June 30, 2019 - August 31, 2019
Assisted tourists visiting Istanbul as part of Fatih Municipality's "Voluntary Tourism Delegates" program.	

Projects

Nano BERT and decoder | *GitHub:* github.com/mrbesher/nanobert, github.com/mrbesher/nanodecoder

A reimplementation of BERT with ALiBi trained on characters as tokens, and an ongoing work to implement a GPT architecture with enhancements like SwiGLU, RoPE, Pre-LN, RMSNorm, multi-head latent attention, and sparse mixture of experts to improve activation, positional encoding, normalization, inference efficiency, and dynamic input processing.

Gaze Tracking Robot | *GitHub:* github.com/mrbesher/gaze-tracking-robot

Developed a Gaze Controlled Robot designed to empower individuals with disabilities to control wheelchairs using eye movements. Used ESP32, OpenCV, MediaPipe, and Tkinter.

The Genetic Art Forger | *GitHub:* github.com/mrbesher/genetic-art-forger

An experiment with genetic algorithms to generate art. The project aimed to create an image starting from the bottom-left corner, without skipping any pixels (analogous to not lifting the "pencil").